Aditya Devarakonda

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Work Experience

Software Engineer Intern — UMBC | Distributed Autograding System December 2024 – May 2026

- Reduced average grading latency from **90s to 54s** during peak submission windows by tuning worker concurrency, queue processing, and grading execution paths across **Python, FastAPI, RabbitMQ, Docker, and PostgreSQL**.
- Reduced incomplete grading jobs by **29%** by designing asynchronous workflows with persistent job-state tracking, idempotent submission handling, retry logic, dead-letter queues, and concurrency controls.
- Built FastAPI, RabbitMQ, PostgreSQL, and Docker-based grading workflows to support high-traffic programming submission windows, enabling reliable job recovery and retryable execution.
- Improved technical supportability for grading latency, worker throughput, failed jobs, sandbox timeouts, and queue backlogs by adding service-level logging, queue metrics, and technical documentation.

Software Developer — Pink Moon Technology February 2023 – May 2024

- Owned requirements for backend and reporting workflows in the team, working with the manager and client stakeholders to translate needs into scoped engineering work, implementation priorities, and delivery requirements.
- Improved **p95 service latency by 32%, from 480ms to 325ms** under load-tested concurrent workloads by optimizing query-processing paths, tuning PostgreSQL indexes, adding cursor-based pagination, and reducing expensive synchronous reporting operations.
- Coordinated dependencies across backend, PostgreSQL, internal APIs, and asynchronous reporting workflows, communicating implementation status, blockers, and requirement changes across the engineering team and stakeholders.
- Reduced failed report jobs by **21%** by designing C++ asynchronous worker workflows using queue-based processing, bounded retries, and failure isolation, while adding structured logs and request tracing to improve root-cause analysis.

Projects

AdTrace — Advertising Measurement Reliability Platform | *TypeScript, React, FastAPI, Playwright, PostgreSQL, OpenTelemetry*

- Built an advertising-measurement reliability platform that reconstructs **browser-to-CRM conversion lineage** and diagnoses failures across consent, click IDs, identity, and offline attribution, achieving 90% root-cause accuracy across 1,000+ held-out scenarios.
- Developed fault-injection, replay, and reconciliation tooling covering **30+ measurement failure classes**, enabling evidence-backed diagnosis of broken partner implementations and conversion discrepancies.
- Engineered deterministic diagnostics, idempotent ingestion, and end-to-end verification across TypeScript/React, FastAPI, PostgreSQL, Playwright, and OpenTelemetry.

Media Manifest Analyzer — Streaming Integration Diagnostics | *Python, HLS, MPEG-DASH, ffprobe, pytest*

- Built a deterministic **Python CLI** for diagnosing HLS and MPEG-DASH streaming integrations, normalizing manifests into a unified rendition and segment model for consistent analysis across formats.
- Implemented diagnostics for missing segments, bitrate-ladder gaps, codec mismatches, discontinuities, target-duration violations, and malformed manifest structures, with optional **ffprobe**-based codec verification.
- Added fault-injection fixtures, pytest-based validation, CLI and JSON output, and actionable root-cause reports to make playback and packaging failures reproducible and easier to troubleshoot.

ApeironCode — AI Engineering Workflow Platform | *TypeScript, Node.js, LLM APIs, LSP*

- Built a local-first AI coding system integrating multiple LLM providers into structured planning, implementation, testing, review, and security workflows with approval-gated filesystem, shell, Git, and network operations.
- Designed persistent project context for goals, plans, tasks, technical decisions, verification results, and execution summaries, enabling multi-step AI workflows to remain traceable and reviewable.
- Evaluated the system across **150 held-out engineering tasks**, achieving **66% verified task success** and **21% fewer regression-causing patches** compared with a single-agent baseline.

Education

University of Maryland, Baltimore County August 2024 – May 2026

Master of Science in Computer Science | **GPA: 3.89/4.00**

SRM University, Amaravati August 2020 – May 2024

Bachelor of Technology in Computer Science | **GPA: 3.70/4.00**

Technical Skills

Data & Analytics: SQL, PostgreSQL, ClickHouse, Data Pipelines, Data Modeling, Query Optimization, Analytics Evaluation, Program Metrics, Metrics & Reporting

AI & Software: Python, C++, Java, JavaScript, TypeScript, FastAPI, Node.js, REST APIs, GenAI / LLM Integration, LangGraph

Program & Execution: Requirements Scoping, Stakeholder Communication, Cross-Functional Coordination, Dependency Tracking, Technical Prioritization, Issue Triage, Project Tracking, Client Communication, Technical Documentation

Infrastructure & Observability: Kafka, RabbitMQ, Redis, Docker, AWS, OpenTelemetry, CloudWatch, GitHub Actions, CI/CD, Structured Logging, Request Tracing